

JEE MAIN REVISION MASTER: LOGARITHMS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Laws of Logarithms & Exponential Equations	2 PYQs	Logarithms

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Laws of Logarithms & Exponential Equations	$\log_b a = \frac{\ln a}{\ln b} = \frac{1}{\log_a b}$, $a^{\log_a x} = x$, $\log_b(x^k) = k \log_b x$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Laws of Logarithms & Exponential Equations (Standard Yield — 2 PYQs)

Core Principle: Laws of logarithms: $\log_b(xy) = \log_b x + \log_b y$; $\log_b(x/y) = \log_b x - \log_b y$; $\log_b(x^k) = k \log_b x$.
Base change rule: $\log_b a = (\log_c a / \log_c b) = (1 / \log_a b)$. Identity: $a^{\log_a x} = x$. Domain of $\log_b x$: $x > 0$, $b > 0$, and $b \neq 1$.

■ **Caution / JEE Pitfall:** Logarithm is undefined for non-positive numbers ($x \leq 0$) and base cannot be 1 ($b \neq 1$).

■ **Speed Trick:** Product property shortcut: $\log_b a \cdot \log_c b \cdot \log_d c = \log_d a$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The value of $e^{\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \log_{10} \tan 3^\circ + \dots + \log_{10} \tan 89^\circ}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?

- (A) The domain of the logarithm function is $\mathbb{R}$, the set of real numbers.
- (B) The range of the logarithm function is $\mathbb{R}^+$, the set of all positive real numbers.
- (C) The point $(1,0)$ is always on the graph of the logarithm function.
- (D) The graph of the logarithm function is decreasing as we move from left to right.

Correct Answer: None

Solution Analysis: ■ Domain $\in \mathbb{R}^+$ and range $\in \mathbb{R}$ and graph of the logarithm function is increasing. So, point $(1,0)$ is always on the graph of logarithm function.